Customer Shopping Behavior Analysis

Data-Driven Insights from 3,900+ Transactions | 18 Features Analyzed | Strategic Business Intelligence

Project Overview

Scope

Comprehensive analysis of customer behavior patterns, including spending habits, segmentation, preferences, and subscription models.

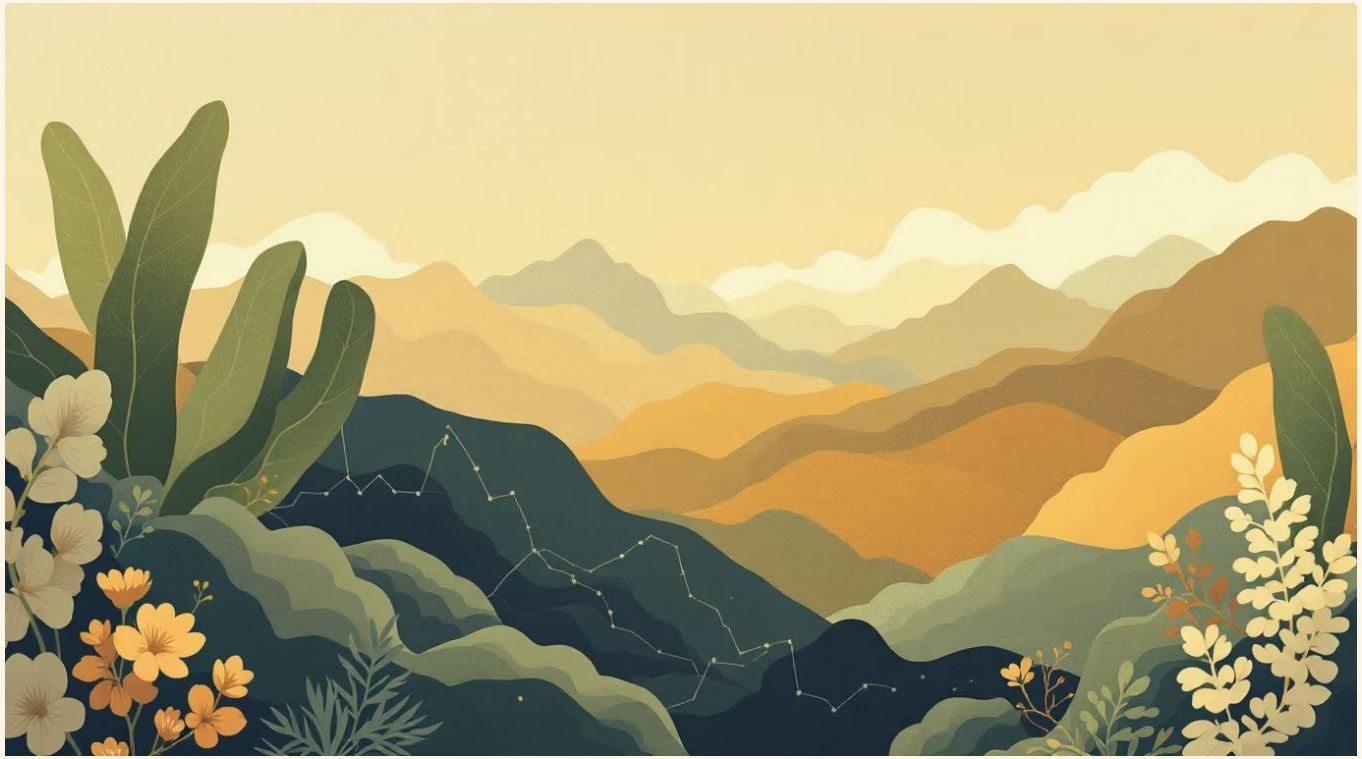
Data

Robust dataset comprising 3,900 records across 18 analytical dimensions, ensuring depth and reliability.

Goal

Deliver actionable intelligence to drive business growth and strategic decision-making.

Dataset Summary



Key Features

Demographics, purchase details, behavior, shipping

Missing Data

37 nulls in Review Rating

Structure

Rows: 3,900 · Columns: 18

EDA — Python Workflow

Load & Inspect

Comprehensive data profiling using pandas: df.info(), .describe()

Clean & Impute

Review Rating imputed by category median using statistical imputation techniques

Engineer Features

Advanced feature creation for predictive modeling: age_group, purchase_frequency_days

DB Integration

Loaded cleaned DataFrame into enterprise-grade PostgreSQL database

SQL Analysis — Key Metrics

Revenue by Gender

Male: 157,890 · Female: 75,191 — segmentation analysis revealing market opportunities

High-Spending Discount Users

839 rows — customers using discounts yet > avg spend, emphasizing customer value optimization and retention strategy

Top Rated Items

Gloves 3.86 · Sandals 3.84 · Boots 3.82 · Hat 3.80 · Skirt 3.78 — product performance analytics and quality insights

Product & Shipping Insights

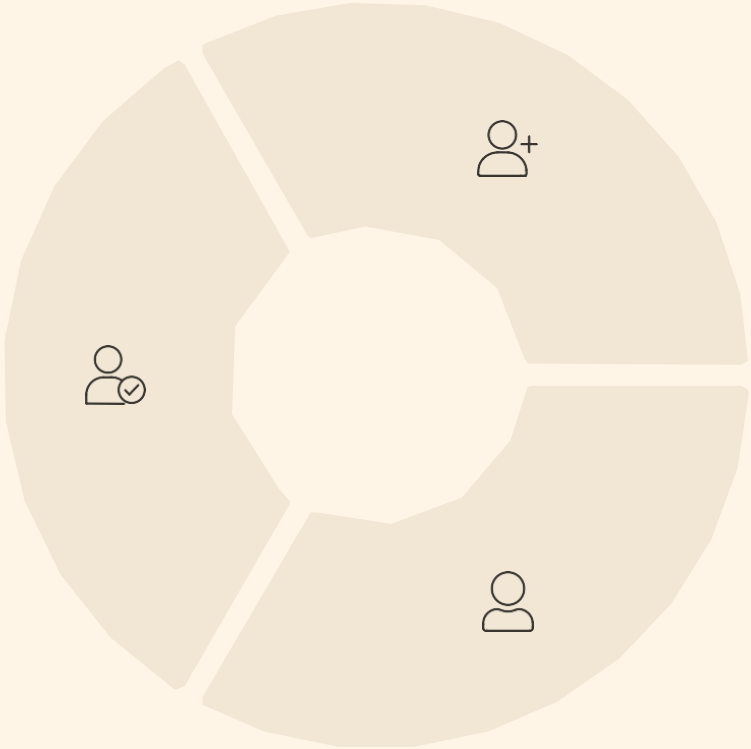
Discount-Dependent

Hat 50% · Sneakers 49.66% · Coat 49.07% · Sweater 48.17% · Pants 47.37%

Shipping Avg Spend

Standard: 58.46 · Express: 60.48

Customer Segmentation



New

83 customers

Loyal

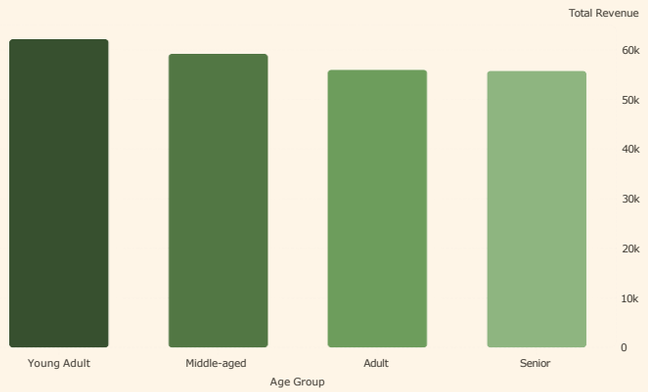
3,116 customers

Returning

701 customers

Revenue by Age Group

Focus marketing on Young Adult & Middle-aged segments — top revenue contributors.



Dashboard Snapshot

Power BI interactive view — quick KPIs

Key metrics: Customers 203 · Avg Rating 3.79 · Avg Purchase ₹61.18 · Subscribers 31.5%

Business Recommendations

Implement subscription monetization strategy with exclusive member benefits

Design tiered loyalty framework to increase customer lifetime value

Optimize promotional strategy to balance growth with profitability

Leverage performance analytics to drive product strategy

Execute data-driven marketing campaigns for high-value segments